Points raised by the reviewer in *italic*. Answers in regular blue text.

1. The Abstract and the Conclusions were not revised in step with the Discussion.
a) The Conclusions (lines 808ff) still state that inertial effects are "crucial for capturing Lagrangian coherent structures and thus crucial when modeling transport and mixing". This is unconditional, while the Discussion tones down the same finding (lines 681ff). The headings of Sections 3.6 and 4.2 have also not been changed. Please revise these three places to be in line with the new discussion. The Abstract is already toned down, but does not include a velocity-scale caveat.

We have revised the Conclusions section to tone down the claims of the importance of the inertial effects in regards to mixing. We have revised the titles of Section 3.6 and 4.2 to better qualify the claim of those subsections. We left the more nuanced discussion of velocity scales to the body of the paper, which we feel is more appropriate given the abstract serves as a concise summary.

b) Same location, "transport and mixing" goes beyond what was computed, since mixing depends on diffusion and no transport equation was solved. The authors could use "advective transport" instead.

We have rephrased from “transport and mixing” to “advective transport” as suggested.

c) Section 4.2 sets bounds to the inertia claim ("Reynolds numbers in the lower hundreds", lines 663ff), but the Reynolds number is not the relevant parameter for this. The unsteady Stokes equations keep the local acceleration and drop only the convective term, so what separates the two models is convective inertia relative to local acceleration, that is the inverse Strouhal number, rather than convective inertia relative to viscosity. At 1 Hz, and with the length scale of about 3.3 mm implied by the stated Re=319 at 6.8 cm/s, the local acceleration exceeds the viscous term by two orders of magnitude, so Re weighs the dropped term against the smaller of the two terms that are kept. Please state the bounds in terms of St. Also, neither St nor Re are defined in the manuscript. Please define both, stating the length scale used and the values at the modelled operating point and at the mm/s velocities discussed in lines 675ff.

We have now introduced and explicitly defined how we compute both Reynolds (Re), Womersley (Wo) and Strouhal (St) numbers in Methods, Sec. 2.7. We have updated our discussion in Sec. 4.2 on the implications of dropping the convective acceleration term when computing Lagrangian coherent structures. Specifically, we have added quantification and consideration of all three above mentioned dimensionless numbers, and how these would be affected by mm/s scale velocities.

2. The statement that the displacement data incorporate any mechanism that possibly deforms the brain (lines 763ff) is new in this revision, but I do not think it is supported. For respiration, for example, this does not hold. The displacement data are cardiac-gated (Almudayni et al. cited by the authors) and gating averages out what is not phase-locked to the heartbeat. The data are then imposed as exactly 1 s-periodic (lines 244ff), so only harmonics of 1 Hz are admitted. Respiration is, therefore, not there at reduced amplitude, but absent as a distinct driver. See, e.g., Sloots et al. (2020, NeuroImage 210:116581),

where the two drivers were assessed individually and found to deform the basal ganglia in opposite directions, cardiac systole expanding the tissue and inspiration compressing it, with the latter attributed to venous outflow at reduced thoracic pressure. Please remove the sentence or change it to where it states that the displacement magnitudes come from in vivo measurements in which other mechanisms were also active. Please also state that the cardiac mechanism is the only time-varying driver in the human model, since secretion is constant in time (lines 778f) and the ciliary traction has constant magnitude (lines 285ff). Please also list what "in some studies" (lines 761ff) refers to. Söderström et al. has now appeared in Magn Reson Med (2026) 96:1916-1928, so the published version should be cited. That study also measured at the foramen magnum, not in the ventricles, in a cohort aged 68 to 82, and reports a respiratory stroke volume of 0.38 mL against 0.77 mL for the cardiac cycle.

We have changed the sentence on other mechanisms to only stating that other mechanisms may be active in the measurements. We have also added to the Methods section a note that the cardiac pulsations is the only time-varying mechanism in our model. Also, we have updated our reference to Söderström et al. and refined our statements in this paragraph.

3. The Results and the Discussion are now at odds regarding the zebrafish compartmentalization.
Lines 621ff state that "the LCS confine CSF to respective ventricles", while lines 655f base the same conclusion on the condition of "a purely advective transport regime", which is preferable. Please align the two. I would also suggest comparing the integration times of 34 to 101 s to the timescale of the omitted transients. Olstad et al. (2019) anaesthetized the larvae to keep bodily movement out of their flow recordings, and report that a single contraction at 2 dpf temporarily eliminates the compartmentalization, with the baseline flow re-established only some 4 to 24 s later. It is thus plausible that the omitted transients could fall inside the analysis window rather than outside of it.

We have aligned the statements of flow compartmentalization in the Results and the Discussion. Additionally, we have added a note discussing the timescales of the transients and development of the LCS we characterized in the Discussion.

4. Siyahhan et al. (2014) appears to be cited for the wrong region (lines 694f).
According to the manuscript, that study showed "only a minor effect of cilia on near-wall flow in the lateral ventricles". However, Siyahhan et al. reports the opposite: cilia raise the time-averaged wall shear stress "by two orders of magnitude from a mean of 0.04 mPa without the cilia up to 2-5 mPa with the cilia", and near-wall dynamics "are dominated by ependymal cilia action". In that study, the region where the bulk flow governs near-wall dynamics is the aqueduct.

This matters because the authors' observation described in the manuscript (lines 692f) is on lobe structures at the aqueduct entrance. The relevant comparison would thus be with the aqueductal result, not the lateral-ventricle one. With that comparison, there would also not be a contradiction with Siyahhan et al. Please correct the attribution and state which region is being compared. At lines 705ff, the inference about the inferior and occipital horns rests on both studies, although neither analyzed the near-wall or Lagrangian flow in those regions; please mark it as an extrapolation.

We thank the reviewer for pointing out the inconsistency in our report of the Siyahhan et al. results. We have corrected the attribution, and have marked the inference about the horns as extrapolation.

Minor points:
Different maximum aqueductal velocities are given in lines 505ff (6.8 cm/s) and lines 543f (7.5 cm/s). To be consistent, the one underlying the stated Re should be used or the difference explained.

Line 543 has been corrected to state 6.8 cm/s.

Lines 245ff state that the magnitudes of "some of the displacements" were reduced, but the scaling factors now given in lines 252ff apply to all five regions, so all of them, not just some, were reduced. Please adjust the wording or the factors.

We have rephrased both line 245 and the preceding sentence, removing the word “some”, since, as the Reviewer points out, all five regions had adjustment factors.

Line 702: "spontaneous muscles"; word missing

Corrected to “spontaneous muscle contractions”.

Lines 424ff report the convergence study and point to Section A.2, but do not mention that maximum displacement magnitude does not converge under mesh refinement (lines 1116f). Please name that exception in the main text.

We have pointed out the exception in the text.

Lines 1085f state that correctness of the discretization schemes was verified on the human brain ventricles geometry using the Method of Manufactured Solutions, but no results of that verification are reported. The two model problems that follow are a channel flow and a Taylor vortex. The ventricular geometry is used only later for consistency checking of certain quantities. Please reword.

We thank the Reviewer for pointing this out. We have updated the Appendix to be more precise in our reporting of the verification.